Colloids and Surfaces A: Physicochemical and Engineering Aspects 726 (2025) 137891

Contents lists available at ScienceDirect

Colloids and Surfaces A: Physicochemical and Engineering Aspects

journal homepage: www.elsevier.com/locate/colsurfa

# Pretreatment approaches to minimize clogging during managed aquifer recharge with drywells

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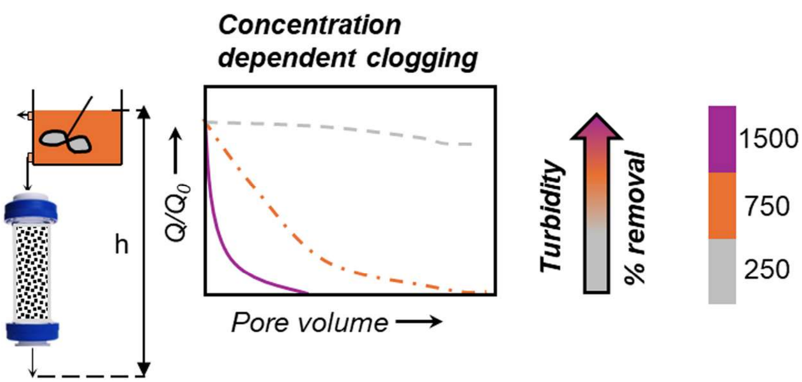
## GRAPHICAL ABSTRACT




## ARTICLE INFO

*Keywords:*
Drywells
Managed aquifer recharge
Porous media
Clogging
Suspended particles

## ABSTRACT

Managed Aquifer Recharge (MAR) using drywells (DWs) is a promising strategy for groundwater replenishment in arid and semi-arid regions. However, clogging caused by suspended particles (SPs) reduces infiltration efficiency and necessitates effective pretreatment strategies. This study evaluates the use of porous media as a sacrificial pretreatment layer—a guard column—to capture SP before they enter the primary infiltration zone. Laboratory column experiments were conducted under a constant head (148 cm) using kaolin clay at turbidity levels of 250, 750, and 1500 FNU over five recharge cycles. Two column configurations were tested: (i) quartz sand alone (15 cm), and (ii) a dual-layer system of quartz sand (9 cm) over black coarse sand (6 cm). To reverse clogging, two unclogging strategies were investigated: (1) a hydraulically optimized backflush, and (2) a modular top-sand replacement approach that involved removing and replacing the upper 3 cm of media. Clogging behavior was concentration (turbidity) dependent. At high turbidity (1500 FNU), straining and hydrodynamic bridging caused severe surface clogging, leading to rapid permeability loss, requiring frequent top-layer replacement. At moderate turbidity (750 FNU), straining and blocking contributed to a gradual permeability decline as retention sites became saturated. Hydrodynamic bridging was also present, but less significant than at the highest turbidity. Backflushing was moderately effective but declined in efficiency over repeated cycles. At low turbidity (250 FNU), fine particles migrated deeper into the porous media rather than accumulating at the surface. Electrostatic repulsion limited attachment, but hydrodynamic forces facilitated deeper

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https://doi.org/10.1016/j.colsurfa.2025.137891
Received 21 May 2025; Received in revised form 22 July 2025; Accepted 28 July 2025
Available online 29 July 2025